

Ena Kovac

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SOFTWARE EXPERIENCES

Software Development Engineer Intern for Amazon AWS

June 2025 – August 2025

- Designed and validated a WebSocket Management Service handling 8 MB payloads and 6 K+ concurrent connections by implementing scalable throttling and persistence mechanisms, and conducting large-scale load testing on AWS ECS
- Reduced manual UI interaction time by 30% by creating MCP Server integrating 38 APIs for natural language automation
- Proposed 3 production-ready solutions through collaborating with senior engineers to analyze key architecture trade-offs

Software Engineering Intern for Samsung SDS (*Research and Market Intelligence Intern*)

June 2024 – August 2024

- Minimized manual update time by 95% through engineering the full automation system for toolkit within ZTM product
- Developed 25+ advanced automation features, including automatic link updates and opt-in/out functionality for toolkit
- Authored a 15+ page manual to ensure smooth handover and guide future maintainers in using and managing the system

Machine Learning Analyst for American Express

August 2023 – June 2024

- Automated dialogue flow classification pipelines using 5 Selenium scripts, improving system reliability by 60% and reducing maintenance overhead through modular architecture, dynamic input handling, and robust error recovery

EDUCATION

Cornell University – College of Engineering

August 2022 – May 2026

- Bachelor of Science in Computer Science, Minor in Artificial Intelligence – Dean's List, GPA: 3.5+

SOFTWARE PROJECTS

Database Development – Java

December 2024

- Architected and deployed a SQL database managing 100K+ records, optimizing large-scale queries for 3 core data models
- Designed custom query operators improving retrieval efficiency by 40% under tight memory and resource constraints
- Created a dynamic query plan builder that automates algorithm selection based on dataset size (1K–100K) and resources

Distributed Messaging System – OCaml

March 2025

- Developed a distributed chat system supporting 1,000+ concurrent clients with real-time, asynchronous communication
- Optimized message routing via socket programming and concurrency, hitting <500 ms latency and 99% message delivery

Emotion Classification Model – Python / NLP / Deep Learning

January 2025

- Engineered and benchmarked an NLP pipeline trained on a 50K-sample dataset, achieving 95% accuracy across 20+ emotion classes by integrating GloVe embeddings, BiLSTM, Attention, CNN layers, and XGBoost with Keras and pandas

LEADERSHIP EXPERIENCES

Vice President of Technology, First Generation Investors

August 2024 – Present

- Developed a website using HTML, CSS, and JavaScript, increasing accessibility and internal member engagement by 50%

Vice President of Technology & Marketing, Epsilon Nu Tau

September 2023 – June 2024

- Engineered a marketing strategy with executive members, boosting online engagement by 45% and recruitment by 30%
- Led startup concept development, conducting market research and effectively pitching business potential to 4 stakeholders

Independent Math Tutoring Business Owner

January 2022 – Present

- Provide personalized one-on-one tutoring in mathematics for students aged 8-20, including those with ASD and ADHD

SKILLS & LANGUAGES

- **Software Languages:** Java, Python, HTML, JavaScript, TypeScript, CSS, OCaml, SQL, C, Assembly, VBA, C++, AMPL
- **Software Tools:** AWS, Docker, GitHub, PostgreSQL, Selenium, TensorFlow, Keras, scikit-learn, pandas, NumPy, Artillery
- **Technical Skills:** Data Structures & Algorithms, Functional & Object-Oriented Programming, Machine Learning, Natural Language Processing, Backend & Web Application Development, Distributed & Parallel Systems, Cloud Computing, Relational Databases, Information Retrieval, Large-Scale System Design, Networking, Scalability, Unix/Linux Environments, Containerization & Microservices, Load Testing & Automation, UI Design, Data Storage, Security